



Modeling the 3D Breast Surface Using Thermography

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Abstract. Breast thermographies can show temperature distribution and detect abnormalities in the body on examination. This work shows an implementation allowing their used for three-dimensional (3d) models of the breast surface from their acquisition in five different angles. Firstly, breast segmentation is performed using U-Nets trained for each angle. Then, the segmented edges are transformed into parametric curves. Geometric transformation converts these curves from two-dimensional (2d) to a three-dimensional after proper rotations and translation of them. These 3D curves are described by B-splines curves and used to build a Rational Non-Uniform B-splines Surface (NURBS) for the breast region. Texture mapping is applied to project the frontal thermal data onto the 3d surface. A Likert scale-based survey was conducted with healthcare experts, and experienced academics to evaluate the final results and compare them with examples of the same persons. The results received high grades of approval and agreements.

Keywords: Thermal imaging · Breast cancer · Three-dimensional models

1 Introduction

Breast cancer is the most common type of cancer among women both in Brazil and globally. As reported by the National Cancer Institute 73,610 new cases of breast cancer were diagnosed in the country last year [1]. Projections indicate that by 2040, this number could increase to over 3 million of new cases [2].

There are many exams to detect breast cancer, mammography is the most popular. Despite, the accumulative effect of the ionizing radiation of this exam, mammography is considered the gold standard [3]. Thermography is a non-invasive exam that analyses the heat emission from the body (breast region in this case). It is affordable, fast, can be used in young and pregnant women, and

presents great possibilities to assist in the detection of diseases [4,5]. A abnormal heat distribution on the images can be related to early cancer formation, because it can increase the metabolism of the area [7].

There are various types of treatments for breast cancer, which can be employed individually or in combination, depending on the stage of the disease and the individual characteristics of each patient. Currently, mastectomy and breast-conserving surgery are the most frequently performed surgical procedures [6]. After surgery to treat breast cancer, breast reconstruction may be an option for some patients. This reconstruction can be done through implants (prosthetic reconstruction) or through the transfer of tissue from other parts of the body (autologous reconstruction) [8].

The use of three-dimensional models can bring significant benefits to the breast reconstruction after cancer cure, promoting more satisfactory outcomes for patients related to the shape similarity. The process of creating these 3D models can be carried out through the utilization of a scanner, which captures detailed information about the three-dimensional coordinates of the breast [9]. However, it is important to mention some disadvantages associated with this approach, such as the high costs involved and the lack of access to this technology in hospitals and clinics, as well as the potential inaccuracy in cases of considerable breast size.

2 Materials and Methods

This study uses thermal imaging from the database published on the Visual Lab website [10,11]. The 640×480 images were obtained with a FLIR SC620 (sensitivity of 0.04°C and range from -40°C to 500°C). This database contains 222 exams, but there are exams unsuitable for the purpose of this work (e.g. those with capture on incorrect angles, mastectomized patients and unclear breast edges), they were removed. After this, 140 exams could be used, each one presenting breast images acquired in 5 direction: frontal (0°), oblique right lateral (45°), total right lateral (90°), oblique left lateral (-45°) and total left lateral (-90°). The temperature matrices from these exams are used. The developed approach employs techniques from two areas of computer vision (CV): image processing (IP) and computer graphics (CG). IP methods are used at the beginning to obtain aspects of the geometry of the breasts from the exams, and in the final step, to map textures corresponding to the temperatures. The intermediate part of the process uses CG methods (or vector imaging) for construction and positioning of curves, definition of transformation matrices and generation of 3-d curves and the surface representing the breast under examination.

2.1 Image Processing Phase

The initial IP phase includes five steps as Fig. 1 shows: preprocessing, segmentation, refinement, identification of the relevant edges and extraction key points for each view.



Fig. 1. Flowchart of image processing steps.

Preprocessing takes the original images (640×480) and resized it to 256×256 pixels. The frontal exam is converted from the original format to RGB (Red, Green, Blue) using the FLIR (Forward Looking Infra-Red) rain palette. In this way, the frontal information is used in the final step for rendering the 3d surface by using texture mapping.

Segmentation uses U-net model of Convolutional Neural Networks (CNN) to performed the breast segmentation [12,13]. In fact, 5 CNNs of this model are trained (i.e. hyper parameters are generated for 5 U-nets) each to be used in one of the 5 exam positions. From the database subset used, 109 (78%) exams were used for training the nets and 31 (22%) were separated for testing. The training used 100 epochs in batches of 2 with 50 epochs. Net weights are initialized with normal distribution of random values. Each convolution uses the rectified linear unit activator (*ReLU*) function to allow passing values that are greater than 0. Adam optimizer with learning rate 1×10^{-4} is used [14]. This allows the region of the breasts to be separated (i.e. segmented) from the rest of the image, as shown in Fig. 2. It is noteworthy to emphasize that, given the preliminary nature of this study, segmentation evaluations in comparison with a ground truth standard were not conducted. The analysis of segmented images was subjectively carried out for validation purposes. Those results without apparently satisfactory segmentations were simply excluded of the process.

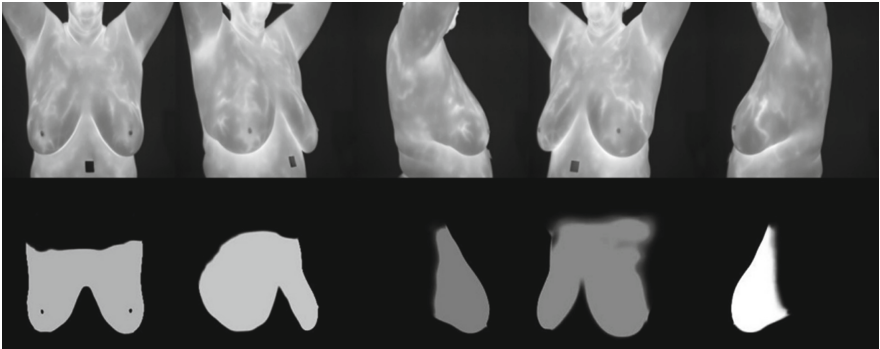


Fig. 2. Example of segmentation outputs with the 5 U-nets.

Refinement was next performed since the U-net segmentation can present more areas than the regions shown in Fig. 2. Such areas must be removed, in order to have only one main region connected per direction. This is done considering the deletion of smaller areas from the output of the previous step. Figure 3 (a) shows an example of a U-net result that leaves smaller areas, they are indicated by the red arrows. For this, a threshold is performed using Suzuki (Fig. 3 (b)) [16], and then only the area with bigger number of pixels is kept in each direction. Finally the Canny's edge detection method (Fig. 3 (c)) is applied resulting only the largest connected edges [15]. This kind of refinement is repeated for each of the 5 views leaving only the largest edge in each one.

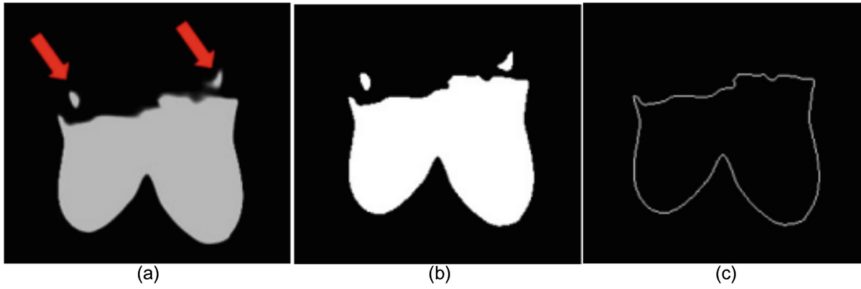


Fig. 3. Refinement of the obtained segmentation in case of more than one area exists as a result of neural network segmentation.

Identification of the relevant edge and points of each view are important because they help in identifying the shape of the breasts, both in width (front) and in thickness and height (sides). They will be used later to generate the 3D surface. The inframammary curves are of main interest, these will be used to adjust all other curves from the relevant edges of each view of the exam in a single 3D coordinate system. To obtain these elements, the identification of the relevant edges uses 4 steps. (1) Initially, the points in the lower region of each edge are found, looking for the first points from bottom to top in each result of the previous step. That is, for each of the final pixels of the horizontal direction, the first pixel of the edge is searched from bottom to top. This will be stored and the subsequent ones ignored. Thus, at the end of this step, only the pixels closest to the bottom of each of the 5 edges from the previous step will be available, as shown in Fig. 4(a). (2) Then, a search is made for the first pixels vertically, starting from the right side of each image. Unlike the bottom-up search, the lateral search consists of a lateral scan of each of the final edge pixels starting at the right end (to find the closest pixels on this side). Output of this step is shown in Fig. 4(b). (3) Same idea is applied to define the edge on the left (to find the closest pixels on this side), this result is shown in Fig. 4(c). (4) Finally, all pixels from step (1) that are found in any of the other 3 detections (steps 2 and

3) are removed. That is, it is deleted from the result of (1) all points also present in the relevant edges on the right and left sides. In this way, the resulting points at the end of step (4) are just the points that are shown in Fig. 4(d).

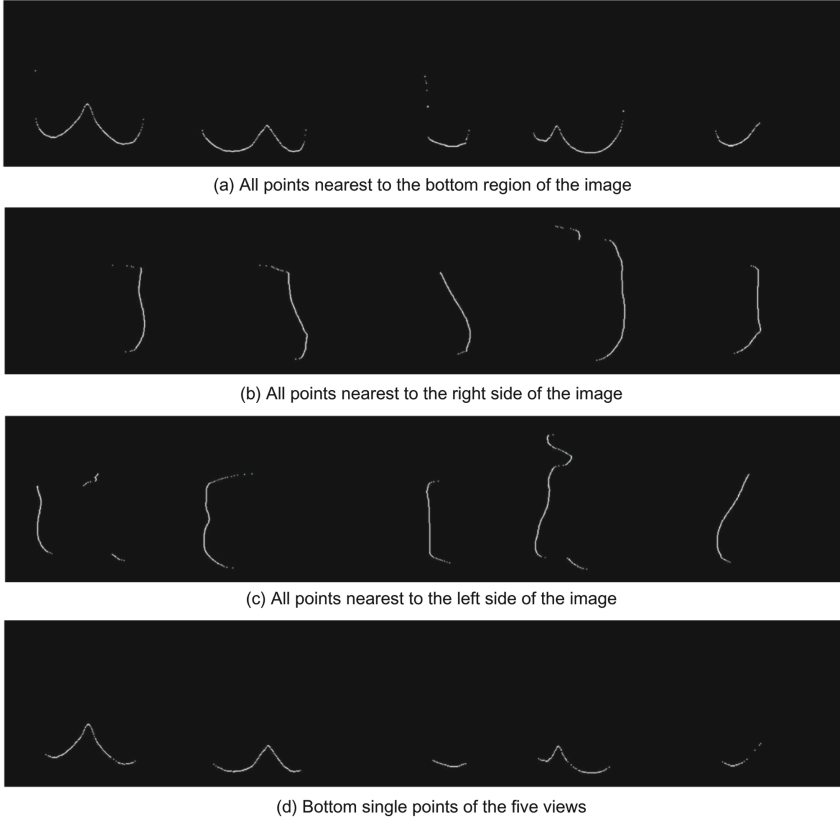


Fig. 4. Identification of unique relevant edges of each view.

Extraction of each view key points for each examination angle (or for each view) must be done, this consists of the identification of three points: P1, P2 and P3 (Fig. 5 show these points). They belong to the lower single edge of the 5 views, that is, from step (4) of the previous subsection. In the 0° , 45° and -45° images, a point is found at the junction of the curves (in the center of the body: P2) and 2 points corresponding to the inflections of the breast curves on the right (P1) and left (P3) first points. For images of 90° and -90° , the beginning and end of the curve are identified, they are presented on Fig. 5(d) (P1 and P3) and the third point is the lowest point of this curve (P2). Figure 5 shows these points in the images resulting from segmentation in order to better identify their meanings.

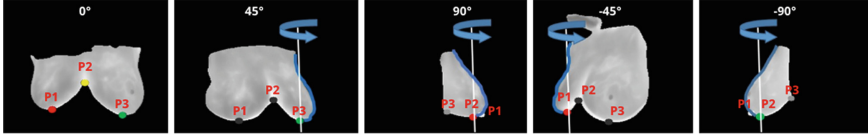


Fig. 5. Identification of the key points of each direction.

2.2 Using Computer Graphics (CG) Techniques

This phase uses CG resources (as shown in Fig. 6) to: (1) transform and place the 2d edges in space, turning them in 9 body curves, (2) adjust the curves by B-splines interpolating them, (3) generate boundary surfaces of the breasts, and (4) render this surface using the frontal thermography of the patient as textures.

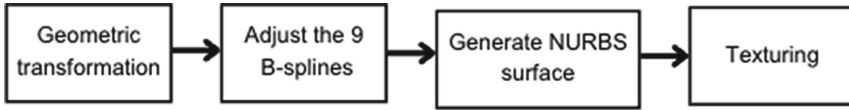


Fig. 6. Flowchart of the steps of the method using CG methods.

To describe the 3D shape of the breast, 9 curves are used (see Fig. 7). From a 0° image, three curves are generated following different criteria: the first curve is defined by all points to the right of P1; the second by all points to the left of P3; and the third is an auxiliary curve that extends from P2 to the top of the image (Fig. 5). Then, four curves represent the complete edges of images rotated at 45° , -90° , -45° , and 90° , respectively. Additionally, two auxiliary curves are created by replicating the edges of 45° and -45° , which will later be rotated at angles of 135° and -135° , respectively (forming the central valley between breasts).

Geometric transformations of the key points needs to use a 3D coordinate system (Fig. 7). The X and Y axis of this coordinate is the same horizontal and vertical directions of the 2d images, respectively, while the Z axis is defined by the vector product of the X by Y. The edge of the 0° image is not rotated. This edge will serve as the base of the 3d model in the z-plane. The edges from the 45° and -90° breast views will have their contours rotated by 45° and -90° , respectively, and positioned over point P3 obtained in the 0° view. The edges from the -45° and 90° breast views will have their edges rotated by -45° and 90° , respectively, and positioned over point P1 obtained in the 0° view. The two auxiliary contours created from copying the 45° and -45° edges will be rotated by 135° and -135° . The processes of translation and rotation are performed through transformation matrices, and all the points of the curves

that pass through these axes are transformed by the corresponding angles. The exclusion of points behind the frontal plane results in the correct final edges. The curves shown in Fig. 7(a) illustrate how the various edge points are positioned in a single view of the breast examination edges. At the end of this process, there are 3D coordinates for all the edge points of the breast in vector form within a single coordinate system.

Adjust of the curves by B-splines from the 9 curves of the previous step. These 3d curves are initially described by all their points (raster format), but in this stage, they will be transformed to vector form, i.e., through mathematical expressions. CG parametric descriptions of B-Splines curves are used for this purpose. Since the number of points in each of these curves is different, a point adjustment by B-spline is used to normalize the number of control points for each curve. The B-spline function implemented in [17] is employed for this purpose. The created curves are of degree 4, and results of definition of the control points that defines the curves obtained in the previous step.

Generation of the NURBS Surface from the 9 B-Splines curves shown in Fig. 7 (a) in blue. The green dots (in Fig. 7 (b)) represent the vertices of the triangles that generated the mesh, obtained after applying a generator of the surface by NURBS [17]. In Fig. 7 (c), the vertices are displayed on a mesh with volume-based coloring, where the bluer the color, the larger the volume, and the redder the color, the smaller the volume, considering a background plane.

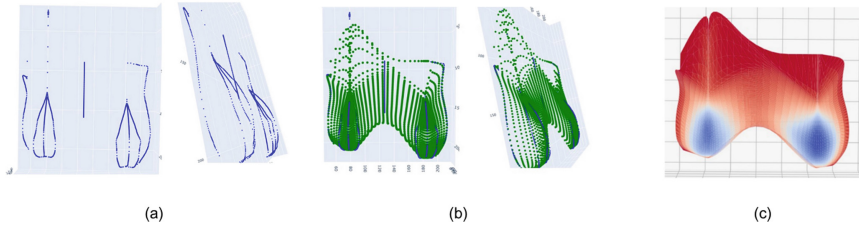


Fig. 7. (a) Points of the 9 curves generated by B-spline for each of the obtained curves seen from two different viewpoints. (b) Vertices obtained after applying NURBS (seen from two different viewpoints). (c) Mesh with volume-based coloring.

2.3 Generation of the Three-Dimensional Thermography

The texture used to render the generated 3-d mesh is produced in the first step (Preprocessing) with false color technique (rain palette of FLIR). Figure 8 shows the texture of the frontal thermographic exam being associated with 2 vertices of the model in a UV texture map. This mapping is repeated for all other vertices and then used to render the region by using OpenGL library [18].

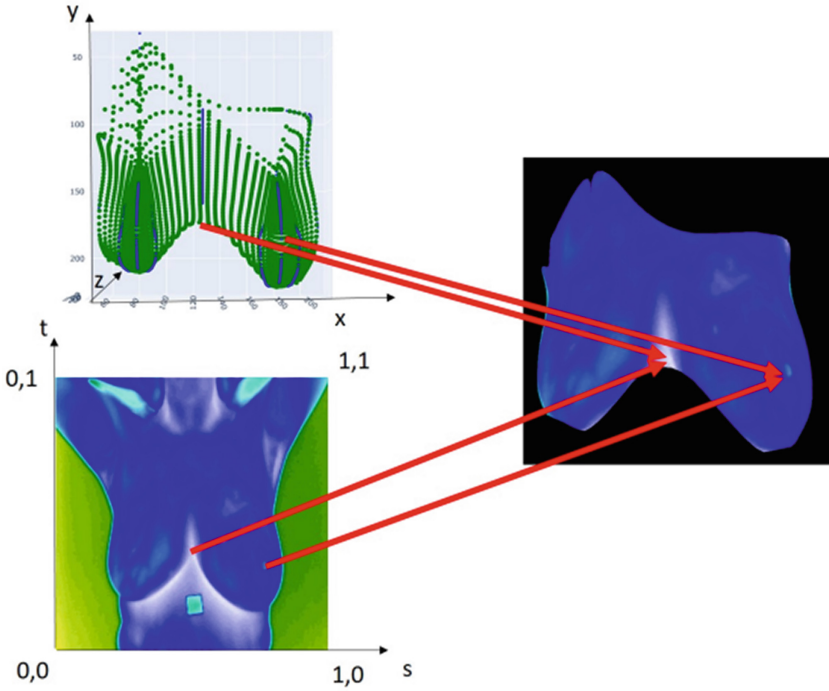


Fig. 8. Final polygonal mesh used for temperature (texture mapping) representation producing a 3d thermogram.

3 Algorithm

All the steps comment in previous section form the complete system that has been implemented using the Python programming language, employing the computer vision library OpenCV for image processing tasks, and using OpenGL for the CG (transformation, curves, surface and rendering) purposes. Algorithm 1 presents a succinct pseudo code representation of the image processing and computer graphics procedure implemented.

Algorithm 1. 3D Surface from 5 Segmentations

```

1: Start
2: Define  $X$ ,  $Y$ ,  $Z$  coordinates ( $X$  horizontal,  $Y$  vertically,  $Z = X \times Y$ )
3: Load segmentations of  $0^\circ$ ,  $45^\circ$ ,  $-45^\circ$ ,  $90^\circ$ , and  $-90^\circ$  exams
4: Load original  $0^\circ$  exam
5: for each segmentation do
6:   Apply threshold  $T$  to binarize it
7:   Perform segmentation refinement using morphological operations
8:   Find and store edges of  $0^\circ$ ,  $45^\circ$ ,  $-45^\circ$ ,  $90^\circ$ , and  $-90^\circ$  segmentations
9:   Find the inframammary curve of each segmentation and name it
      2d_list_of_relevant_edges
10: for  $0^\circ$ ,  $45^\circ$ , and  $-45^\circ$  2d_list_of_relevant_edges do
11:   Find the highest point  $(x, y)$  and name it  $P2$ 
12:   Find the lowest point
13:   if lowest point is the right side of the curve then
14:     name it  $P1$ 
15:   if lowest point is the left side of the curve then
16:     name it  $P3$ 
17: for  $90^\circ$  and  $-90^\circ$  2d_list_of_relevant_edges do
18:   Find the lowest point and name it  $P2$ 
19:   Find the edges end point in direction of the body center and name it  $P1$ 
20:   Find the edges end point in direction of the arms and name it  $P3$ 
21: for each 2d_list_of_relevant_edges (define the 3d curved) do do
22:   Translate each  $P2$  to  $(0, 0, 0)$ 
23:   Rotate the relevant edges angle
24:   if relevant edges is 45 or -90 then
25:     Translate it to  $P3$  of  $0^\circ$  relevant edges
26:   if relevant edges is -45 or 90 then
27:     Translate it to  $P1$  of  $0^\circ$  relevant edges
28: Create auxiliary  $45^\circ$  edge
29: Rotate  $45^\circ$  relevant edges of  $135^\circ$ 
30: Translate it to  $P3$  of  $0^\circ$  relevant edges
31: Create auxiliary  $-45^\circ$  edge
32: Rotate  $-45^\circ$  relevant edges of  $-135^\circ$ 
33: Translate it to  $P1$  of  $0^\circ$  relevant edges
34: Define body edges
35: Select point of  $0^\circ$  segmentation at left of  $P1$  and name it Left body edge with
       $Z = 0$ 
36: Select point of  $0^\circ$  segmentation at right of  $P3$  and name it Right body edge with
       $Z = 0$ 
37: Create a line from  $(P2_x, P2_y, 0)$  of  $0^\circ$  segmentation to  $(P2_x, 0, 0)$  and name it
      middle line
38: Remove all points with positive  $Z$  coordinates
39: Represent each 3D curve as a B-Spline
40: Use control points of the 9 Splines to create a NURBS
41: Store these 3D control points
42: Include texture on the NURBS using as UV map the original  $0^\circ$  exam
43: Show the result
44: End

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4 Results

This article presents an system to model the body from 5 thermograms and also a way to produce a 3D visualization for thermographic examinations. The methodology created in original and uses a combination of IP and CG techniques. The first part of this methodology extracts the shape of the breasts based on their edges and then the second part transform these edges in a 3d curves that generates a spatial model using NURBS. The algorithm used in the system after the segmentation is present, as well. 140 exams was used generating the same numbers of 3D models and exams. The generated models can be viewed at [19]. Figure 9 shows the result for the volunteer with ID 143 of the data set bellow the original acquisitions for visual comparison.

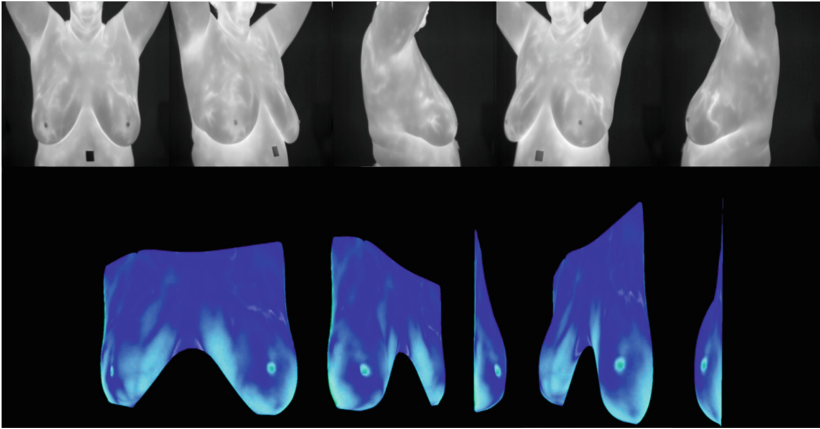


Fig. 9. Thermographic images of volunteer ID 143 and she 3D reconstruction by this technique.

The 3D resultant models were visually evaluated by comparing the exams and the results in a survey using a five-point Likert scale. Fifteen experts in the fields of healthcare, computer graphics, 3D modeling, and image processing, all with experience in medical image analysis, were consulted regarding 10 of the created 3D models. The results achieved an approval rate of over 70%, which is considered a very good outcome [20].

In the complete implementation, 5 U-nets Convolutional Neural Networks (CNNs) were employed for the segmentation of breast regions. As a future work, considering that the right and left lateral images are mirrored images, the segmentation process can be optimized by vertically flipping the images, thus training a single network to handle both lateral and oblique views. This approach reduces the complexity of the model and simplifies the breast segmentation process. Another enhancement would be the use of only one network to

segment the breast in all views, improving the algorithm's efficiency and yielding accurate results [21,22]. In future works, we are considering also to compare the generated models with real 3-d models acquired through a 3D scanner and reconstructed using other methods to verify better the accuracy of the presented method through numerical quantifiers such as average distances or coincidence of breast relevant points. In other words, a comparative study of the models constructed using this methodology with real 3-d models will be conducted on the same day of the thermogram acquisition. Thus, the patient's breasts will be captured in 3D to enable a numerical comparison between the generated models and the real body. This will help validate the methodology, assess its accuracy, and evaluate its clinical utility.

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